

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specifications for Pig Trap End Closure
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Specification for Pig Trap End Closure

R01	22.06.22	Issued for Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	22-06-22	First Issue for Clients Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-012

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-006	Pig Trap Calculation Note
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe

EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011	Specification for Pig Traps
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
ANP-CW212280-ASNP2-LA-2105-00008	Data Sheet for Pig Traps
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets 1 to 8
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-010	Datasheet for Pig Traps Enclosure
EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001	Pipeline Design & Operating Philosophy

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BOD	Basis of Design
DEP	Design and Engineering Practice
FEED	Front End Engineering Design
ID	Internal Diameter
ISO	International Standard Organisation
MFD	Manifold
OD	Outer Diameter
OML	Oil Mining Lease
Q.O.C	Quick Opening Closure
ROW	Right of Way

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

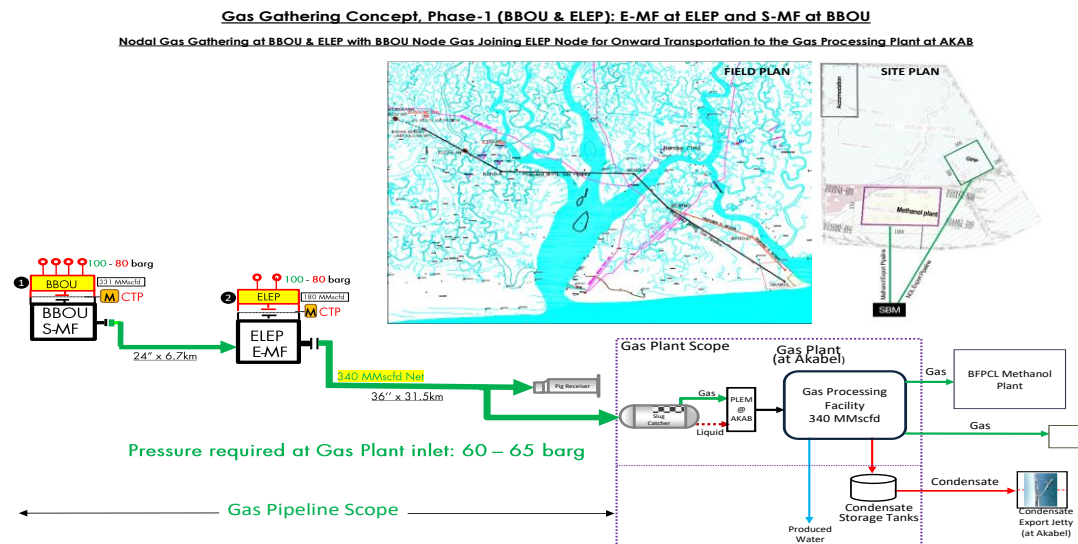


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to specify the minimum requirements for the design, fabrication, testing and inspection of the Quick Opening Closures (QOCs) and the Valve Interlock Systems of the Pig Traps (Pig Launcher/Receiver) that shall be installed at BBOU, ELEPA and AKABEL stations for the Pipeline systems of the BFPCL Gas Gathering Project-Phase 1.

This document shall be updated during detailed engineering when the process requirement for valve interlocking system is fully established and shown on the Process flow sheets.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.24	Piping – Engineering and Layout Requirements
DEP 31.40.10.13-Gen (Feb. 2019)	Design of Pipeline Pig Trap Systems
API 5L	Specification for Line Pipe
ASME B31.8	Gas Transmissions and Distribution Piping Systems
ASME B16-47	Large Diameter Steel Flanges NPS 26 Though NPS 60
ASME Sec. VIII div.1 & DIV. 2	Boiler And Pressure Vessel code
ASME Sec. V	Non-Destructive Examination
ASME Sec. IX	Pressure Vessels Welding and Brazing Qualifications
ANSI B 16.5	Steel Pipe flanges and Flanged Valves and Fittings
ANSI B 16.9	Factory-Made Wrought Steel Butt-welding Fittings
ANSI B 16.21	Non-Metallic Flat Gaskets for pipe Flanges
ASTM A 694	Standards specifications for forging carbon and alloy steel, for pipe flanges, fittings, valves and parts for high pressure transmission service
ASTM A 350	Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings and valves and Parts for High T. Service
ASTM A 516	Standard specification for pressure vessels plates, carbon steel, for moderate and lower temperature service
ASTM A 333	Standard specification for seamless and welded steel pipe for low temperature service
ASTM A 193	Alloy Steel and Stainless-Steel Bolting Materials for High T. Service

ASTM A 194	Carbon and Alloy Steel Nuts for Bolts for High Pressure and High T. Service
ASTM A 240	Heat Resisting Chromium and Chromium Nickel Stainless Steel Plates, Sheet and Strips for Fusion – Welded Unfired Pressure Vessels
EN 10204	Material Certification
ISO 9001	Quality Management System

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Environmental and Pipeline Design Data

3.1. Environmental Data

Refer for the basis of design document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

3.2. Pipeline Design Data

3.2.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.2.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Ultimate Tensile Strength	570MPa
OD / Wall Thickness	23.82mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

4. Responsibilities and Scope Of Supply

4.1. Scope

Scope of work includes but not limited to design, materials, fabrication, inspection and testing, surface preparation and painting, documentation, preparation for shipment, preservation, and supply of Quick Opening Closures (QOCs) and the Key Interlock Safeguarding Systems as specified in the datasheets, and specification as listed herein.

The Pig traps under the Gas gathering - Phase 1 project are follows;

Sn.	Size (in)		Pig Trap Type	Location	Pipeline System.
	Major Barrel	Minor Barrel			
1	28"	24"	Pig Launcher	BBOU manifold station	24" BBOU to Elepa Pipeline
2	28"	24"	Pig Receiver	Elepa Manifold Station	24" Pipeline from BBOU
3	40"	36"	Pig Launcher	Elepa Manifold Station	36" Elepa to Akabel Pipeline
4	40"	36"	Pig Receiver	Akabel Pigging Station	36" Pipeline from Elepa

As a minimum, the supply shall include but not limited to the following:

- Quick Opening Closures
- Key Interlocking systems for QOCs and the Valves
- Davit / hinge arrangement for the quick opening closure
- Special tools required for operation of quick opening closure
- All accessories required for efficient functional operation
- All mechanical design in accordance with relevant codes and standards

4.2. Scope of Service

VENDOR's scope of services shall include but not be limited to:

- Fabrication, inspection, and testing
- ASME U-Stamp for QOC
- Mechanical design of quick opening closure
- Mechanical design of valve interlock systems
- Preparation of all documentation and submission for review & approval
- External and internal surface preparation, external painting, preservation
- Spare parts for 2 years operation, all certification, and statutory approvals (if any).

VENDOR shall make all possible efforts to comply strictly with the requirements of this specification,

international codes and standards, datasheet, drawings, and all other requirements as listed therein.

No deviation or exception from this specification shall be permitted without the written approval of the OWNER. Intended deviations shall be listed by VENDOR supported by reasons for consideration by the OWNER. In the absence of any exceptions, it will be presumed that VENDOR fully complies with the specifications and its attachments.

Compliance with this specification shall not relieve VENDOR of the responsibility of furnishing equipment and accessories of proper fabrication and workmanship to meet the applicable international codes & standards, OWNER's requirements and other contractual obligations.

4.3. Exclusions

- Installation at site/Erection.
- Commissioning

5. Technical Requirements

5.1. General

The equipment shall be designed for a life of 30 years, during which, correct and regular preventative maintenance in accordance with the Vendor's manual, shall be carefully and judiciously followed.

A Pig Trap is a piece of pipeline equipment that allows easy loading or unloading of a pig into, or out of, the pipeline. Its purpose is to provide in a safe manner and without flow interruption the means to either insert and launch a pig into the pipeline or receive and retrieve a pig from a pipeline.

A pig trap system comprises the following basic components:

- A quick opening closure, which provides quick and easy access to the major barrel when open and seals the bore when closed.
- A major diameter section, referred to as the Major Barrel.
- An eccentric reducer for connecting the major and minor barrels.
- A minor diameter section corresponding to the line pipe size and referred to as the minor barrel.
- Various nozzles such as vent, drain, pressure indicator, kicker, bypass, relief, equalizing and pig signaller.
- Lifting lugs, supports and earthing lugs
- Valves – vent line valves, kicker line valves, drain line valves, isolating valves

5.2. Quick Opening Closures

Quick Opening Closure shall be designed, fabricated, and stamped in accordance with **UG 35.2** of ASME Section VIII Division 1, and other design requirements contained in the documentations referenced in section 2 herein.

The quick opening closures shall be equipped with a safety system that allows opening it only with the equipment has been totally pressure relieved. Safety interlock device shall be designed to make it impossible to disengage the closure operating mechanism while there is pressure remaining in the equipment.

The quick opening closures shall be designed for vertical installation. The closure opening configuration shall be specified in the datasheet.

VENDOR shall equip the closures with protective weather covers to protect the door and its mechanisms from harsh environmental conditions.

VENDOR shall utilize standard seal materials (Nitrile, Viton, HNBR) for the closure or any approved equivalent.

All moving parts of the closures shall have lubrication facilities.

The minimum allowable swing opening angle for closure shall be 135°.

The quick acting design should allow opening and closing by one man in a period of approximately one minute, without the use of additional devices. Closures 18" and larger shall be hand wheel operated. Large size heavy closures shall be fitted with mechanical means to break seal when opening door and compress seal when closing.

In particular, a fail-safe design of the opening mechanism shall ensure that the failure of any part of the opening mechanism shall leave the closure closed rather than open.

Closure shall be **Bandlock-2** or **TD Williamson type**, or approved equivalent by OWNER.

5.3. Interlock Safeguarding Systems

Interlock systems shall be designed, fabricated, tested and inspected as per manufacturer's standard.

These key interlock system as an additional safety feature shall be provided on the quick opening closure and the valves.

The key interlock system shall be capable of operating on a coded key transfer principle to control the sequence in which the Pig trap valves can be opened or closed.

The following valves/devices (as a minimum) shall be equipped with the interlocking system of the Pig Traps:

- Quick Opening Closures
- By-Pass line valves
- Kicker line valves
- Vent line valves
- Drain line valves
- Two main isolation valves at pig trap inlet

VENDOR shall provide sequence control unit for the valves equipped with interlocking system for the Pig Traps system.

The sequence of valve operation shall be defined in a specific procedure for both launching and receiving pig traps by Vendor.

The procedure shall ensure that the pig trap door cannot be opened under pressure and that the pig cannot be launched without all valves being operated in the correct sequence.

6. Materials

Materials for fabrication of the Quick Opening Closures and the Valve Interlocks shall conform to ASME specification as specified in the datasheet for Quick Opening Closures and Valve Interlocks. Substitution of material requires OWNER approval prior to start of fabrication. All materials shall be new, free from mill scale, defects and imperfections.

The material of the QOC end for pig traps shall be compatible for welding with the material of the pig trap major barrel.

All materials used in the fabrication shall be provided with material certificates for approval stating the following:

- Mill identification
- Material standard and grade
- Heat identification
- Chemical composition by ladle and product analysis
- Mechanical test results
- Other relevant information

The materials for Pig Trap System shall conform as specified below:

- Major and Minor barrel to be API 5L Gr. X70 (wall thicknesses to be calculated by Vendor)
- End closures: ASTM A 694 Grade F65
- Eccentric Reducer: ANSI B16.9
- Flanges: ANSI B16.5
- Valve Interlock (Material to be advised by VENDOR)

All materials to be used in the construction of pig traps shall be procured from OWNER approved Vendors.

7. Manufacturing and Fabrication

Manufacturing of the pressure components: shall be in accordance with DEP 31.40.10.13-Gen. and ASME VIII div 1. Welding assembly of the Quick Opening Closure (Under ASME VIII div. 1) shall be covered by the ASME IX

The VENDOR shall submit to OWNER for approval, all detailed shop drawings, design calculations and all welding procedures and qualification records prior to commencement of fabrication.

The following requirements shall be met as a minimum;

- Minimum spacing of circumferential welds between centerlines shall not be less than four times the pipe wall thickness or 4", whichever is greater.
- Flange bolt holes shall straddle the horizontal and vertical centerlines.
- Welding procedure and qualification records shall be in accordance with ASME 1104.

8. Shop Testing and Inspection

8.1. Inspection

All controls shall be in conformity with codes, norms, standards and requirements indicated in this specification or other documents in the Purchase Order.

Vendor shall supply at the beginning of the job, its own inspection plan with times and list of operations, checks and tests.

Vendor shall give four (4) weeks' notice in writing, the date and place where equipment will be available for verifications, inspections and tests. OWNER or its representative reserves the right to witness by their inspectors.

No surfaces shall be painted before completion of all inspections and without prior authorization of OWNER.

Vendor shall submit to OWNER or to its representative, all materials certification, radiographs and any other data necessary to verify compliance with the requirements of this specification.

Vendor shall make available to OWNER's inspectors all equipment, instruments, utilities and any other form of assistance necessary to check the work.

All tests and inspections shall be at VENDOR's expense.

All modifications for contract conformity of the supply shall be at Vendor expense.

8.2. Shop Tests

All tests shall be carried out by the VENDOR in conformity with the procedures beforehand established by him and in conformity with codes, norms & standards.

These tests and checks shall be as a minimum the following:

Material certificates including chemical analysis, mechanical tests, and hardness tests required for EN 10204 3.2 certification

- Checks during manufacturing:
- Material identification & thickness checks.
 - Welding and weld repair.
 - Non-destructive examinations (radiographic, ultrasonic, liquid penetrated, magnetic particle test)
 - Hydrostatic test
- Cleanliness and drying after hydrotest,
- Final verifications (overall dimensions, customer's connections, nameplate data, painting,

documentation & certification).

- Preparation for shipment and shipping documents.

A functional test of the Quick Opening Closures shall be performed at workshop.

9. Painting/Protection and Spare Parts

9.1. Painting/Protection

After testing and inspection, the external surface of equipment shall be shop painted in accordance with OWNER specifications and project Painting and Coating for Aboveground Specification.

Internal surface shall be protected by a temporary corrosion protection product (min DFT of 30 microns).

9.2. Spare Parts for START-UP

The Vendor shall provide a list of the spare parts to be envisaged for start-up according to his experience.

9.3. Operational Spare Parts

The definition of spare parts is attached to the guaranty of operation continuity and is correlated to the following parameters:

- Wear of components,
- Probability of trouble,
- Operating conditions, environmental conditions included,
- Availability and time of procurement for the parts themselves,

Operating time to be indicated in the design specification.

The Vendor will submit a list of the spare parts recommended for the exercise period of two years, indicating type and quantity. The list will be issued on the OWNER's forms. They will be supplied with desiccators in the packaging for long conservation.

10. Equipment / Material Identification

The QOC shall have a permanently attached, impression stamped, 316 stainless steel name plate.

The nameplate shall include the following information:

- Description
- Tag No.
- Size
- Purchase Order No.
- Design pressure and temperature
- Date of fabrication
- Service
- Corrosion allowance
- Fabrication and Testing Codes
- Third Party Inspectors Name

11. Expediting

OWNER or his representative shall have the right to examine progress during the production of the units of the equipment to verify whether progress is in accordance with the schedule agreed upon and whether the work is being executed in accordance with the provisions stated in this document.

At the request of OWNER or its representative, VENDOR shall inform OWNER or the said representative of all matters relevant to the progress of the work executed by Him or any sub-contractor employed by the Vendor.

12. Preparation For Shipment, Packing and Transportation

Equipment shall be prepared for export shipment and shall be encased in a wooden crate to provide maximum protection during shipment and extended outdoor storage. Following shop testing, equipment shall be drained, evacuated, dehumidified, and sealed. Items which may be damaged during shipment shall be removed, packed separately and shipped with the unit.

VENDOR shall provide OWNER with all the instructions necessary to preserve the integrity of the equipment during storage, at job site and before start up.

Tools and spare parts shall be packed in metal boxes suitable for long term storage. The boxes shall be galvanized or coated to prevent corrosion of tools.

13. Quality Assurance

The Supplier and any Sub-Supplier shall operate a quality system satisfying the applicable ISO 9001 series or equivalent standard provision, appropriate to the goods and services provided.

The OWNER reserves the right to carry out a quality and technical review at both Supplier's and Sub-Supplier's plants.